

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine if the sequence converges or diverges. If it converges, find the limit.

a) $a_n = \frac{\ln(n^2)}{\ln(n+1)}$

b) $a_n = \frac{(-1)^n n^2}{n^3 + 1}$

c) $a_n = \cos(n)$

d) $a_n = \tan\left(\frac{\pi n}{4n+1}\right)$

e) $a_n = \frac{\sin(2n)}{1+\sqrt{n}}$

f) $a_n = \frac{n!}{(2n)!}$

g) $a_n = \frac{-(4^n) + e^{-n}}{2^{2n}}$

2. Find the limit of the sequence $\left\{ \sqrt{5}, \sqrt{\sqrt{5}}, \sqrt{\sqrt{\sqrt{5}}}, \dots \right\}$

3. Kyle is trying to determine whether the sequence $a_n = \frac{2n-3}{3n+4}$ claims that since the denominator of the fraction will always be larger than the numerator, that the sequence decreases. Show, using all 3 methods discussed in the notes, just how wrong Kyle is.

4. Determine if the following sequences are increasing, decreasing, or neither. Also determine if the sequence is bounded.

a) $b_n = \frac{1}{n} - n$

b) $c_n = (-2)^{n+1}$

5. A sequence a_n with $a_n > 0$ for all n satisfies $\frac{a_{n+1} - a_n}{a_{n+1} + a_n} = \frac{1}{2}$. Suppose $a_1 = 2$.

a) Determine the value of a_5 ?

b) Determine if the sequence a_n converges or diverges.

6. Suppose the sequence $\{a_n\}$ satisfies $\lim_{n \rightarrow \infty} (2n-1)a_n = 16$

a) What is the value of $\lim_{n \rightarrow \infty} a_n$?

b) What is the value of $\lim_{n \rightarrow \infty} na_n$?

7. Suppose the sequence $\{a_n\}$ follows the recursion $a_{n+1}^2 = 2a_n + 3$ with $a_1 = 7$. It can be shown that a_n is decreasing. Determine $\lim_{n \rightarrow \infty} a_n$.